COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 2 : SLO – 1 | Operational concepts
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section A – Match the Following (Answers)

- 1. MAR → C. Holds addresses for memory access
- 2. MDR → D. Temporary storage of data
- 3. IR → A. Holds the instruction currently being executed
- 4. PC → B. Points to the next instruction
- 5. General-purpose Registers → E. Scratch-pad registers for holding data and intermediate results

Section B – Fill in the Blanks

- The PROCESSOR (CPU) executes tasks by following a sequence of instructions.
- The MAR register holds the address of the memory location to be accessed.
- The instruction LOAD loads data from a memory location into a register.
- The MDR register stores data being transferred between memory and processor.
- STORE instructions overwrite the content of the destination register or memory location.

Section C – Activity Task

- Instruction: Load R2, LOC | Operation: reads the contents of memory location LOC and copies them into register R2 | Involved: PC, MAR, MDR, memory location LOC, register R2 | Outcome: R2 ← [LOC]; memory content unchanged (read is non-destructive).
- Instruction: Add LOCA, R0 | Operation: adds the contents of memory location LOCA to register R0 | Involved: MAR, MDR, memory location LOCA, register R0, ALU | Outcome: R0 ← R0 + [LOCA]; LOCA unchanged, R0 overwritten with the sum.

Section D – True or False

- The Program Counter (PC) holds the data that is currently being executed. — FALSE (it holds the address of the next instruction)
- The Memory Address Register (MAR) holds the instruction being fetched from memory. — FALSE (it holds the address; MDR holds the instruction/data)
- The contents of R2 and R3 are altered during the execution of Add R4, R2, R3. — FALSE (only R4 changes)
- Load instructions preserve the original contents of the memory location. — TRUE
- General-purpose registers can be used to store intermediate results of operations. — TRUE